Xi'an Jiaotong-Liverpool University

西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT202		INTELLIGENT SCIENCE	

2nd SEMESTER 2024/25 RESIT EXAMINATION

UNDERGRADUATE – YEAR 3

COMPLEXITY OF ALGORITHMS

TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

1. **This is a closed-book examination.**
2. **Total marks available are 100. This exam will count for 100% of the final mark.**
3. **Answer all questions. There is NO penalty for providing a wrong answer.**
4. **Only English solutions are accepted. Answer should be written in the answer booklet(s) provided.**
5. **All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.**

Question 1. [3+8+14=25 MARKS]

A) Compute the following expressions that are given in postfix notation. Show the procedure.

5 7 * 3 4 1 + * + **[3 marks]**

B) What is the asymptotic upper value of the expression $\sum_{i=1}^{n} i^2$ as a function of n by using the Big-O notation? Justify your answer. **[8 marks]**

C) Give an upper tight bound of the runtime complexity class for each of the following two code fragments in Big-O notation, in terms of the variable N. Justify your answers.

1) **[7 marks]**

```
int sum = 0;
for (int i = 1; i <= N - 5; i++) {
   for (int j = 1; j <= N - 5; j = j * 2) {
     sum++;
   }
}
```

2) **[7 marks]**

```
int sum = N;
for (int i = 0; i < 1000; i++) {
   for (int j = 1; j <= i; j++) {
      sum += N;}
   for (int j = 1; j <= i; j++) {
      sum += N;}
   for (int j = 1; j <= i; j++) {
      sum += N;}
}
```


Question 2. [9 MARKS]

Given two 64 bit integers a and n, here is an algorithm to compute a^n

```
Power(a,n):
1. If n = 0: return 1.
2. If n = 1: return a.
3. If n is even:
4. Return Power(a×a,n/2).
5. else:
6. Return a×Power(a×a,(n−1)/2).
```

Assume throughout this problem that we do not need to worry about overflow (a^n fits into a 64-bit integer variable) and that each operation on a 64-bit integer takes O(1) time.

1) Let A(n) denote the running time of Power(a,n). Write a recurrence relation for A(n). **[3 marks]**

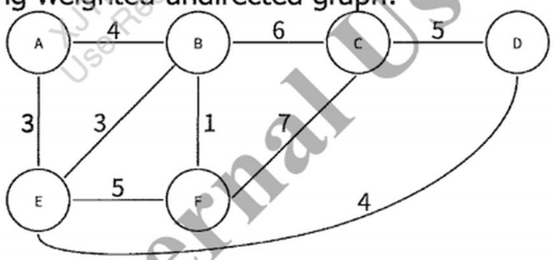
2) What is the solution (in Big-Theta notation) to your recurrence A(n)? Justify your answer. **[6 marks]**

Question 3. [16 MARKS]

Insert the following sequence of elements into an AVL tree, starting with an empty tree: 10, 20, 15, 25, 30, 16, 18, 19. Draw the AVL tree after each insertion. **[16 marks]**

Question 4. [10+5=15 MARKS]

Given the following weighted undirected graph:



(1) Show the step by step process of using the Prim's algorithm to find the minimum spanning tree. We start with node A. And draw the minimum spanning tree. **[10 marks]**

(2) What is the total weight of the minimum spanning tree? **[5 marks]**

Question 5. [8+5+7=20 MARKS]

Alice and Bob are using the RSA cryptosystem for secure communication. Bob has chosen two prime numbers p=5 and q=17 to generate his public and private keys.
(1) Compute n, Φ(n), and select a valid encryption key e (e is less than 10). **[8 marks]**

(2) Encrypt the message 4 according to the selected encryption key e. **[5 marks]**

(3) Derive the private key d. **[7 marks]**

Question 6. [5+10=15 MARKS]

Let G=(V,E) be a directed graph. Let S be a subset of edges of G such that deletion of S (edges) results in a graph G' with no directed cycles. Such an S is a feedback arc set. The size of S is the number of edges in S. The Feedback Arc Set (FAS) decision problem is "to determine for a given integer k if G has a feedback arc set of size at most k".

(1) Is the Feedback Arc Set (FAS) problem in NP? Prove or disprove it. **[5 marks]**

(2) Provide a reduction from the Vertex Cover (VC) problem to the Feedback Arc Set (FAS) problem, and prove its correctness. **[10 marks]**

END OF EXAM PAPER